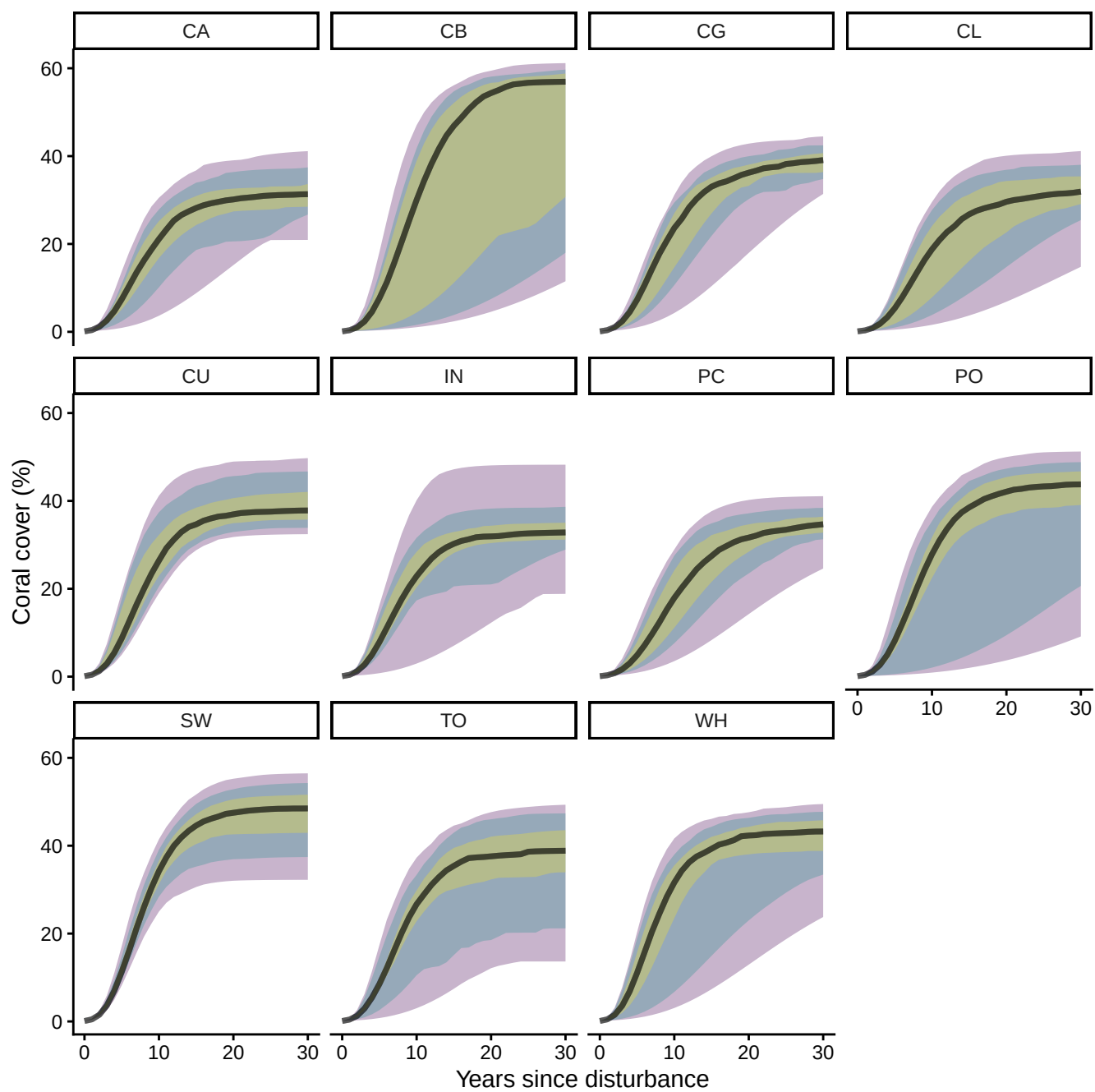
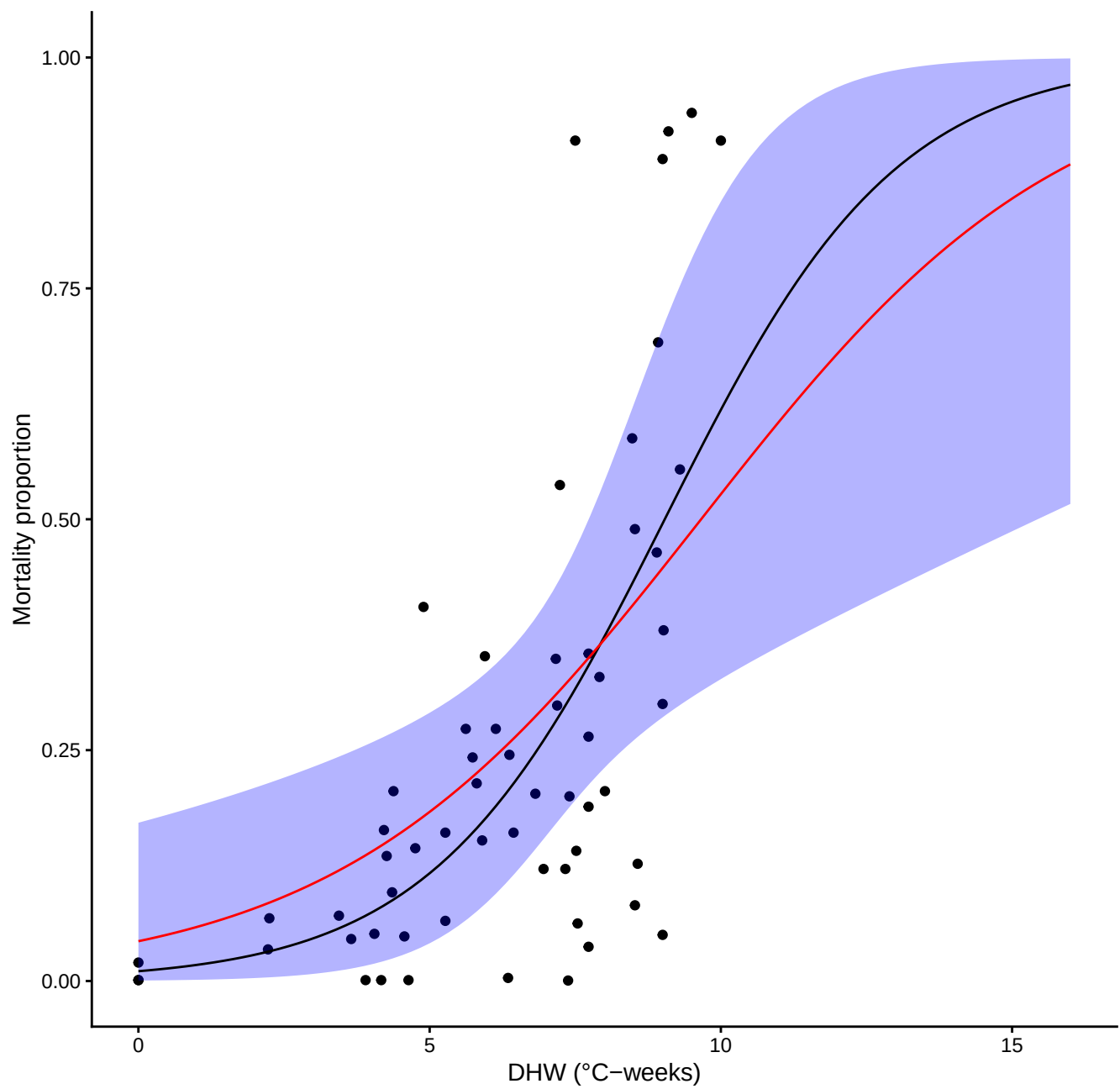
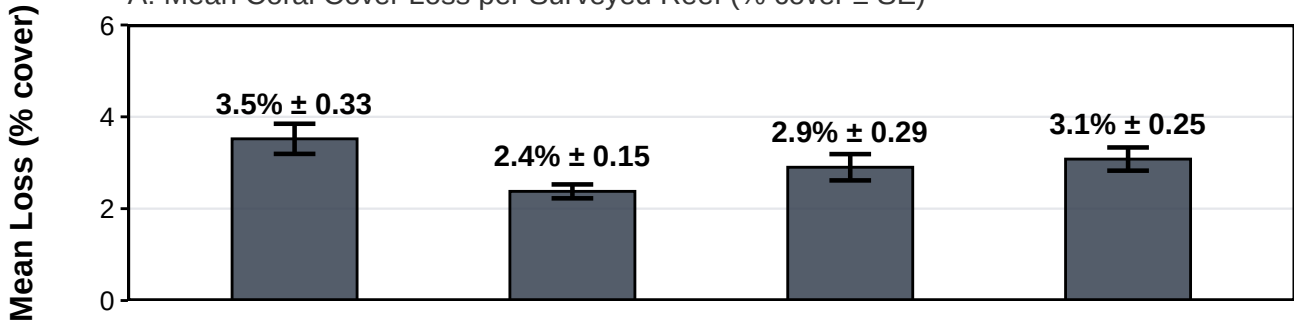




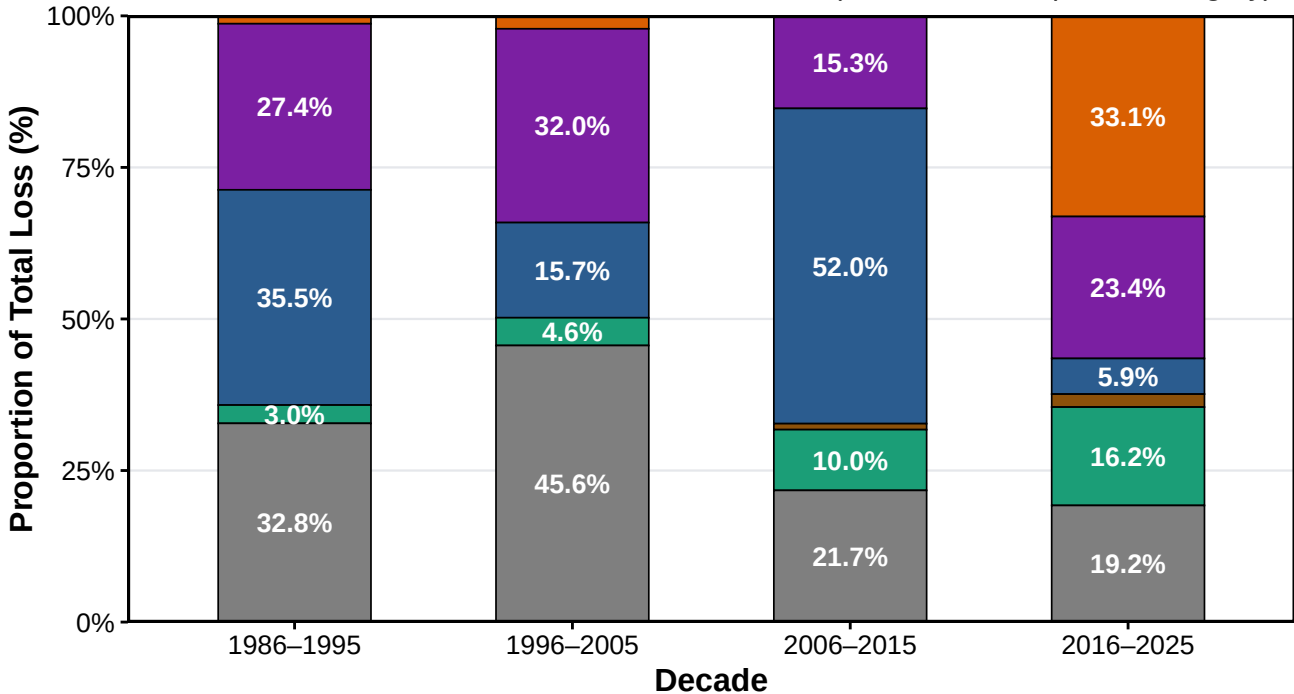


Coral Disturbance Attribution Across Four Decades (1986–2025)

A: Mean Coral Cover Loss per Surveyed Reef (% cover $\pm$ SE)



B: 100% Stacked Bar Chart of Disturbance Attribution (Plot 1: 'm' as Separate Category)

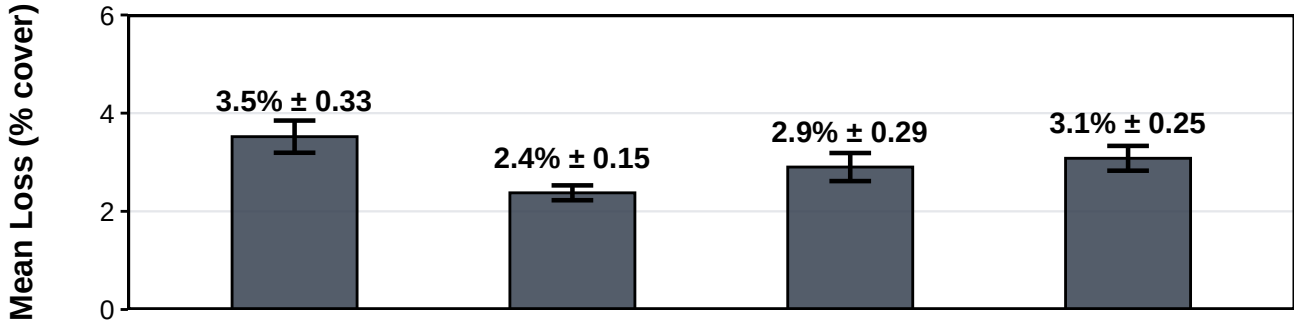


Disturbance Type

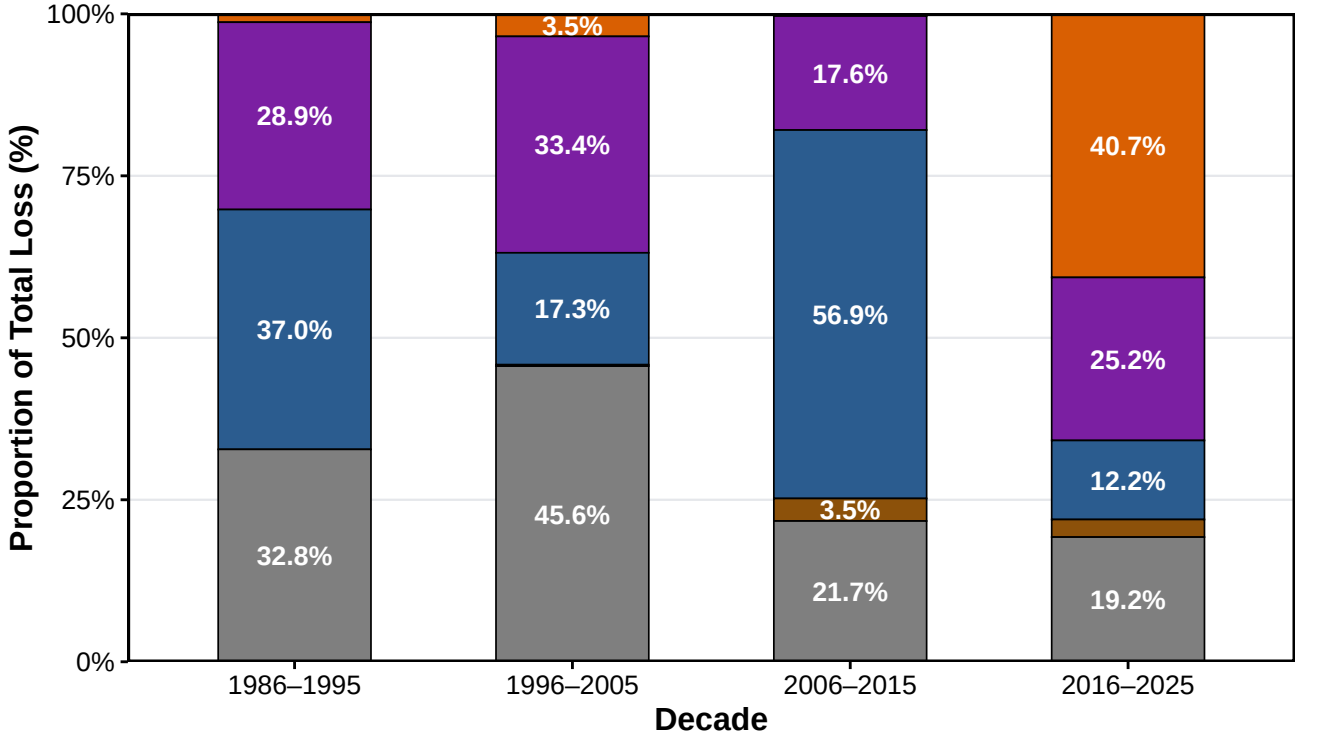
- Bleaching
- COTS
- Cyclone/Storm
- Disease
- Multiple (m)
- Unknown/Other

Coral Disturbance Attribution Across Four Decades (1986–2025)

A: Mean Coral Cover Loss per Surveyed Reef (% cover $\pm$ SE)



B: 100% Stacked Bar Chart of Disturbance Attribution (Plot 2: 'm' Partitioned)



Disturbance Type



Bleaching



COTS



Cyclone/Storm



Disease

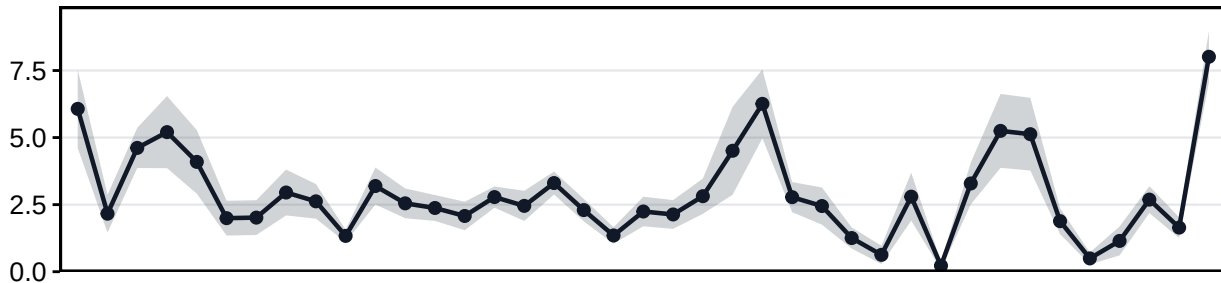


Unknown/Other

Annual Coral Cover Loss and Disturbance Attribution (1987–2026)

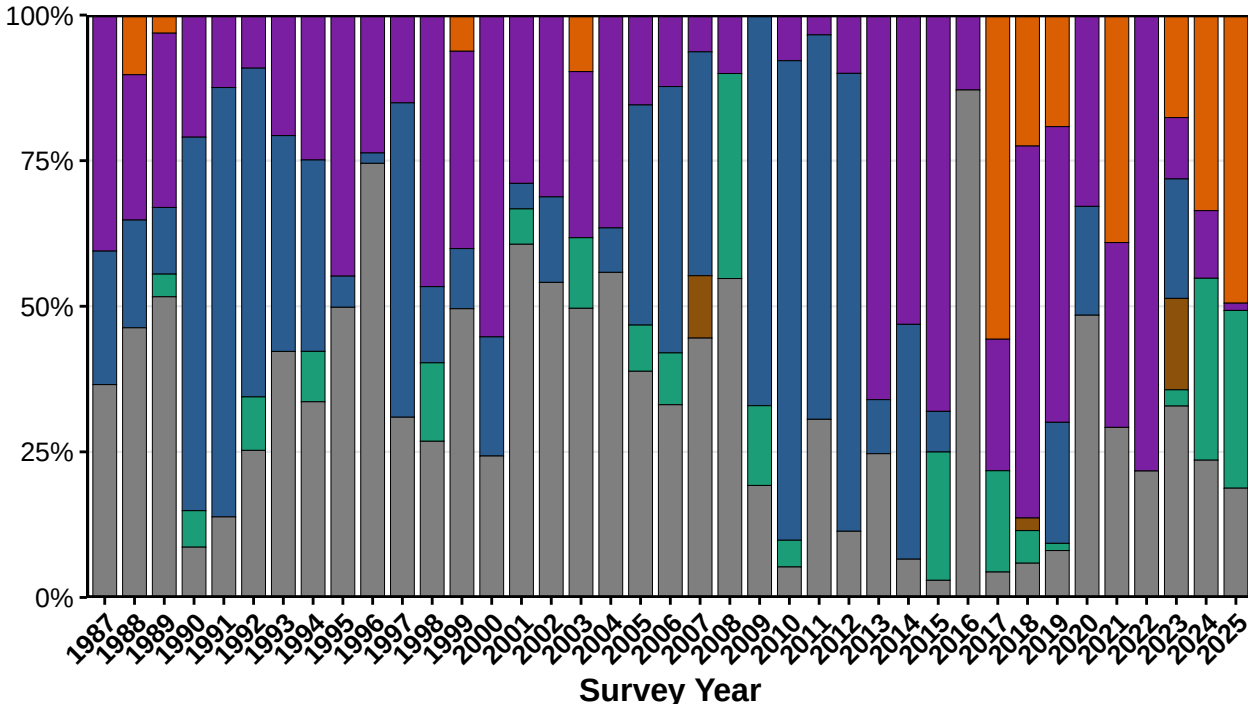
A: Mean Hard Coral Cover Loss per Surveyed Reef (% cover $\pm$ SE)

Mean Loss (% cover)

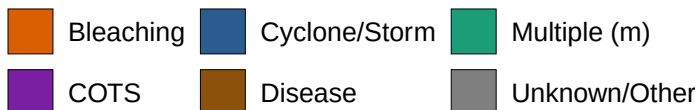


B: Annual 100% Stacked Bar Chart (Plot 1: 'm' as Separate Category)

Proportion of Total Loss (%)

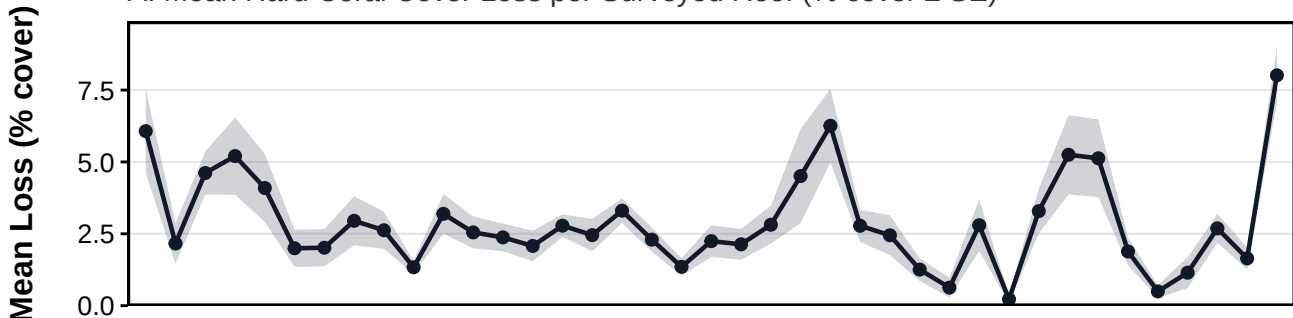


Disturbance Type

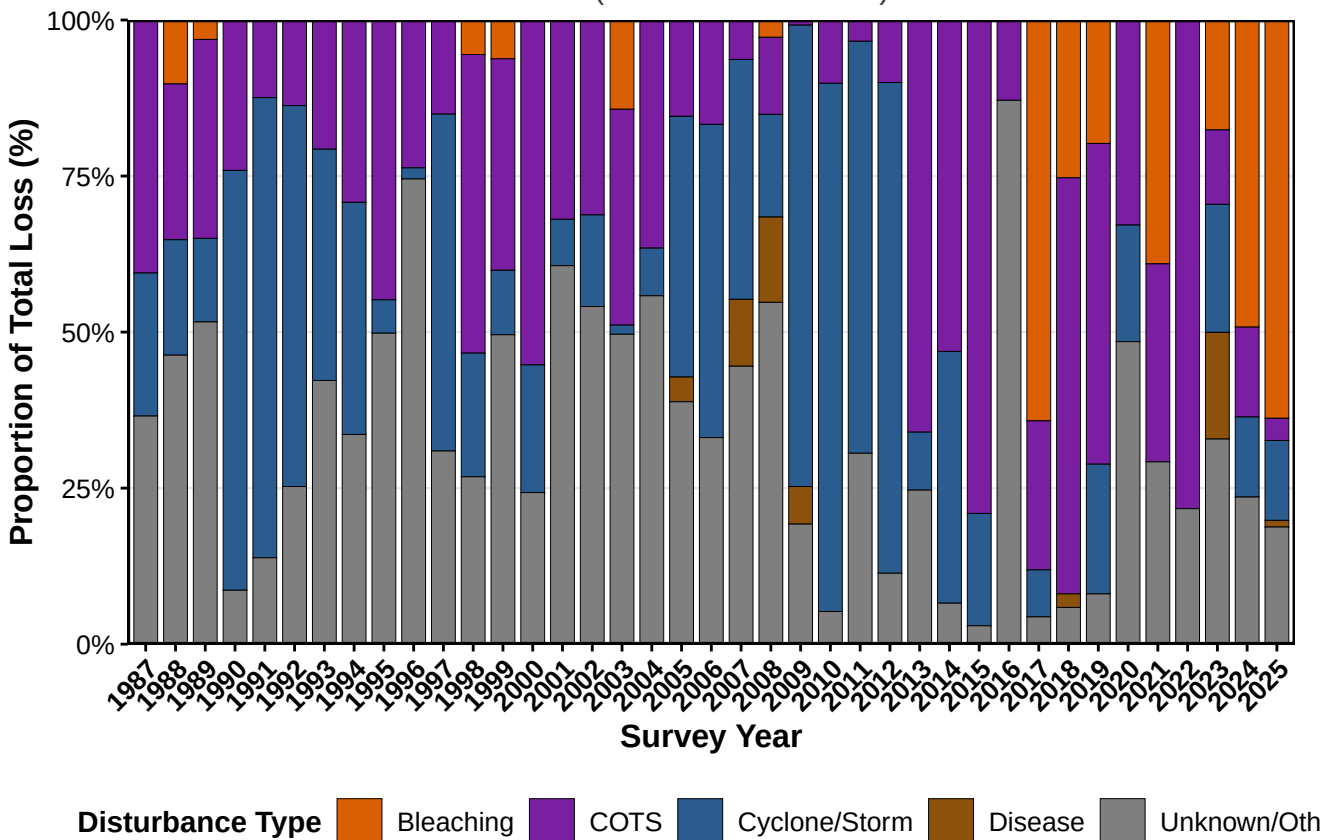


Annual Coral Cover Loss and Disturbance Attribution (1987–2026)

A: Mean Hard Coral Cover Loss per Surveyed Reef (% cover $\pm$ SE)



B: Annual 100% Stacked Bar Chart (Plot 2: 'm' Partitioned)



Coral Trajectory & Disturbance Attribution: Overall GBR & Regional

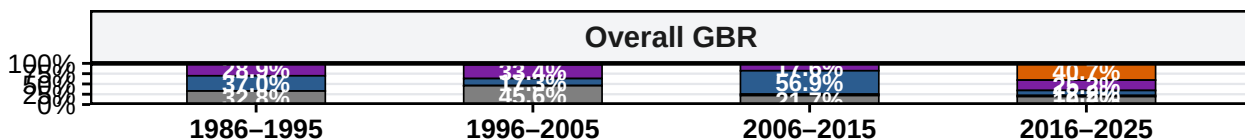
A: Overall GBR Mean Net Coral Cover Change per Surveyed Reef (% cover $\pm$ SE)



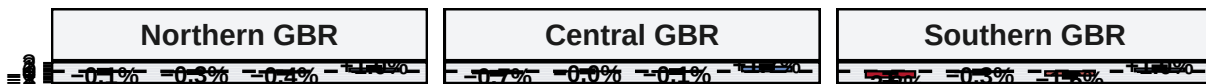
B: Overall GBR Mean Hard Coral Cover Loss per Surveyed Reef (% cover $\pm$ SE)



C: Overall GBR 100% Stacked Bar Chart ('m' Partitioned)



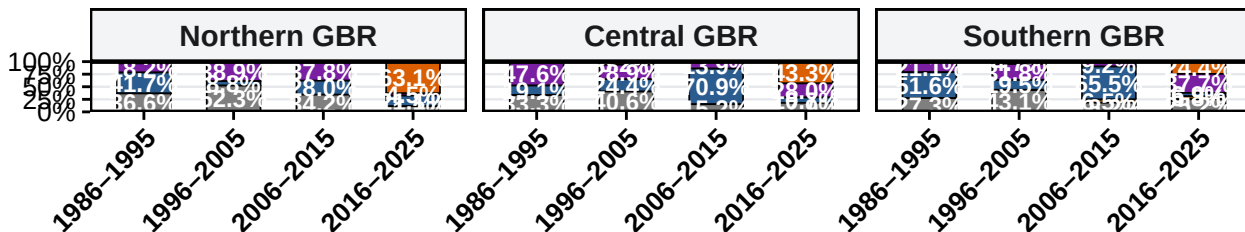
D: Regional Mean Net Coral Cover Change per Surveyed Reef (% cover $\pm$ SE)



E: Regional Mean Hard Coral Cover Loss per Surveyed Reef (% cover $\pm$ SE)



F: Regional 100% Stacked Bar Charts ('m' Partitioned)



Disturbance Type



Bleaching



COTS



Cyclone/Storm



Disease



Unknown/Other

Annual Coral Trajectory and Disturbance Attribution (1987–2026)

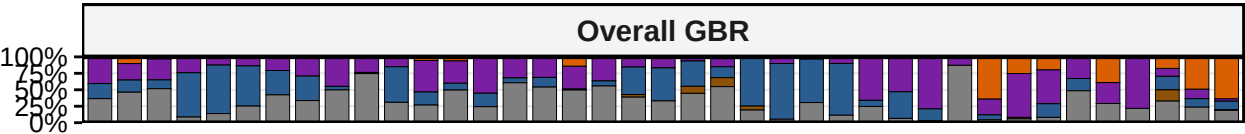
A: Overall GBR Mean Net Coral Cover Change per Surveyed Reef (% cover $\pm$ SE)



B: Overall GBR Mean Hard Coral Cover Loss per Surveyed Reef (% cover $\pm$ SE)



C: Overall GBR Annual 100% Stacked Bar Chart ('m' Partitioned)



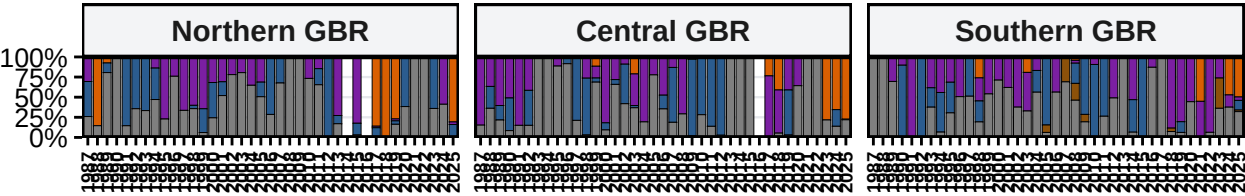
D: Regional Mean Net Coral Cover Change per Surveyed Reef (% cover $\pm$ SE)



E: Regional Mean Hard Coral Cover Loss per Surveyed Reef (% cover $\pm$ SE)

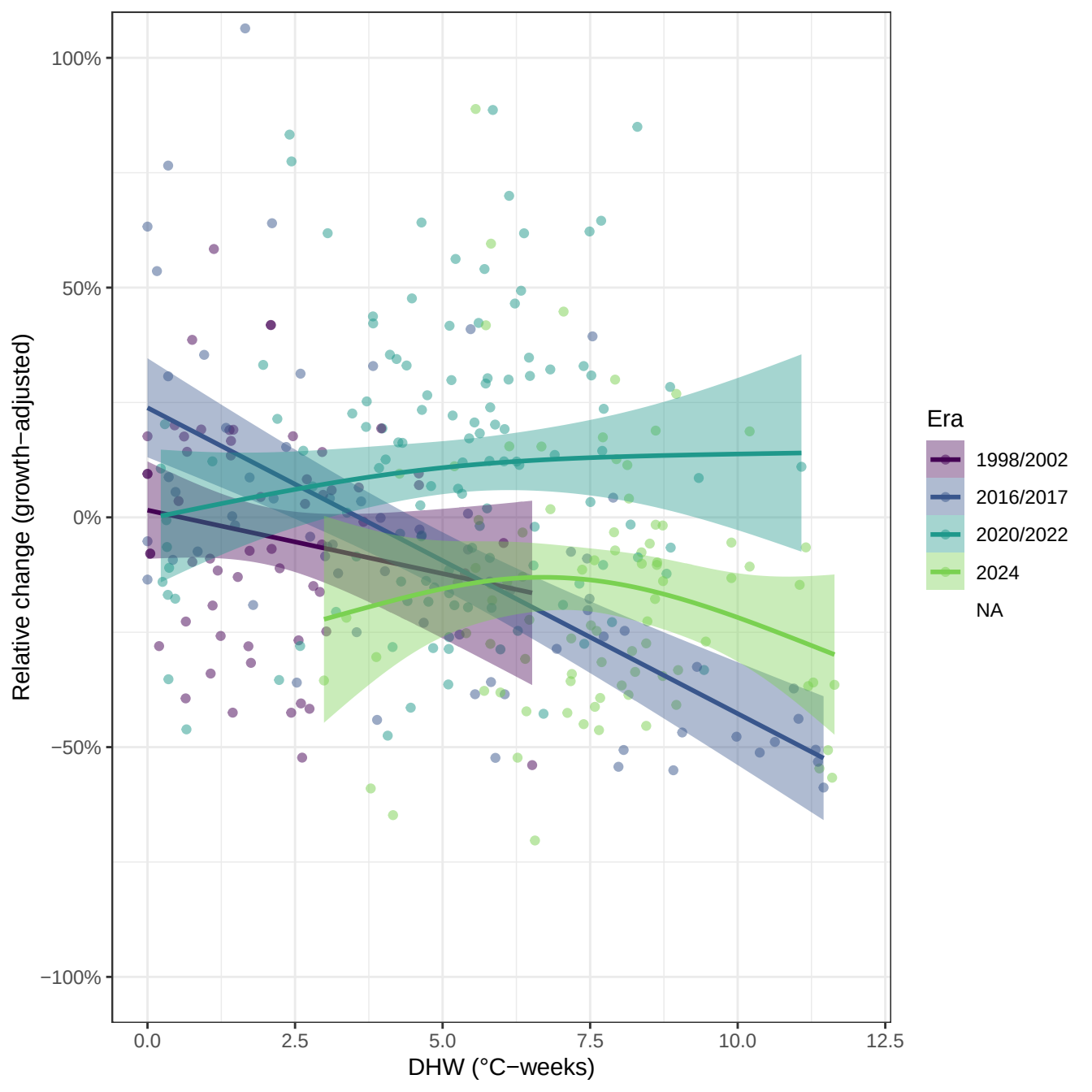


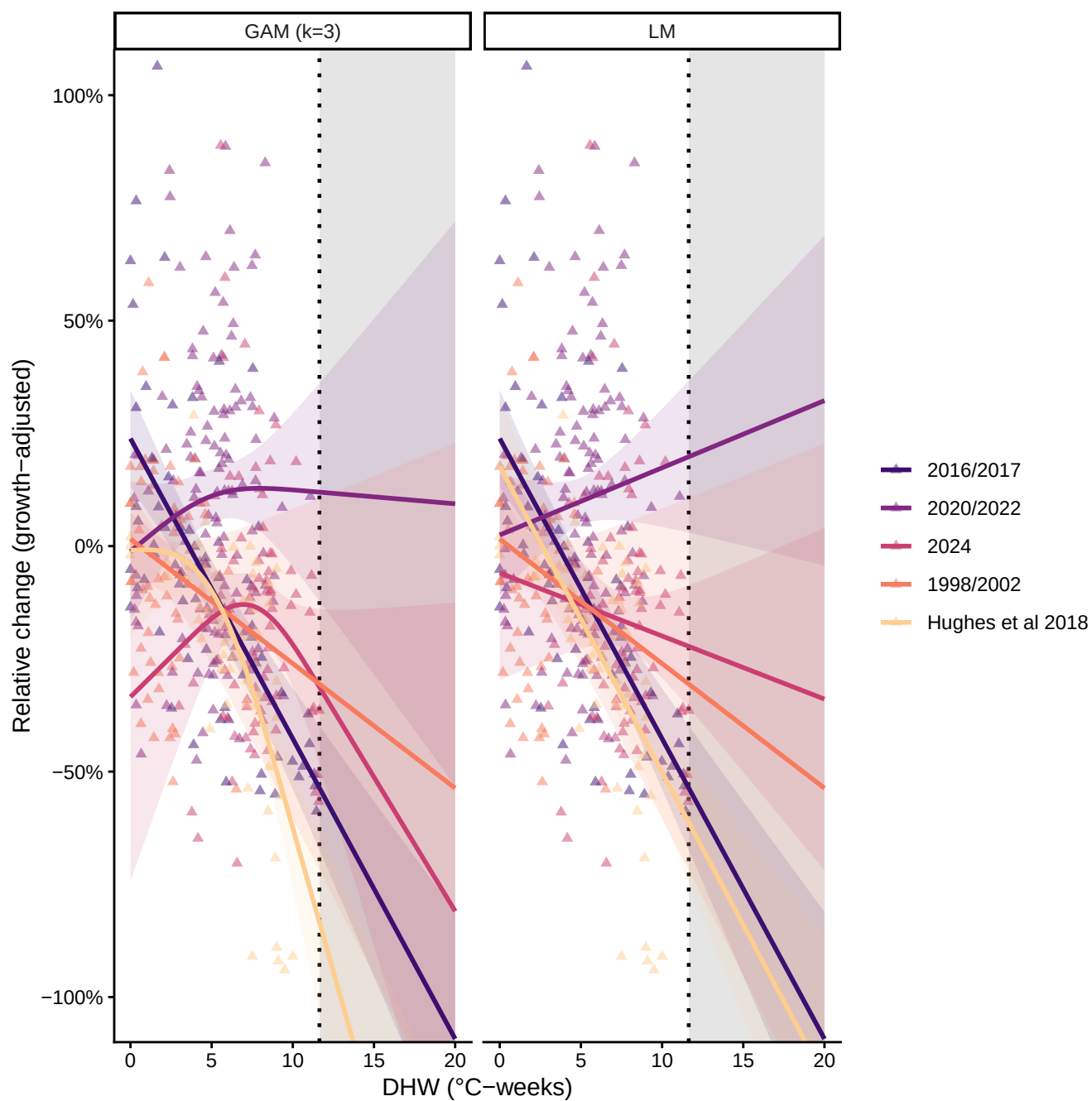
F: Regional Annual 100% Stacked Bar Charts ('m' Partitioned)



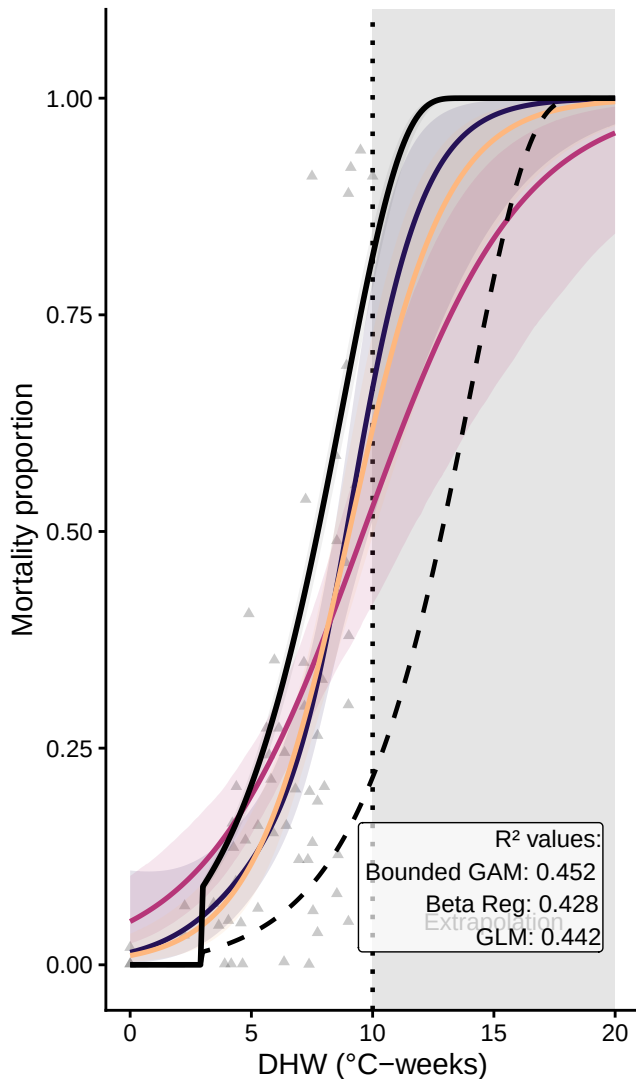
Survey Year

Disturbance Type Bleaching COTS Cyclone/Storm Disease Unknown/Other

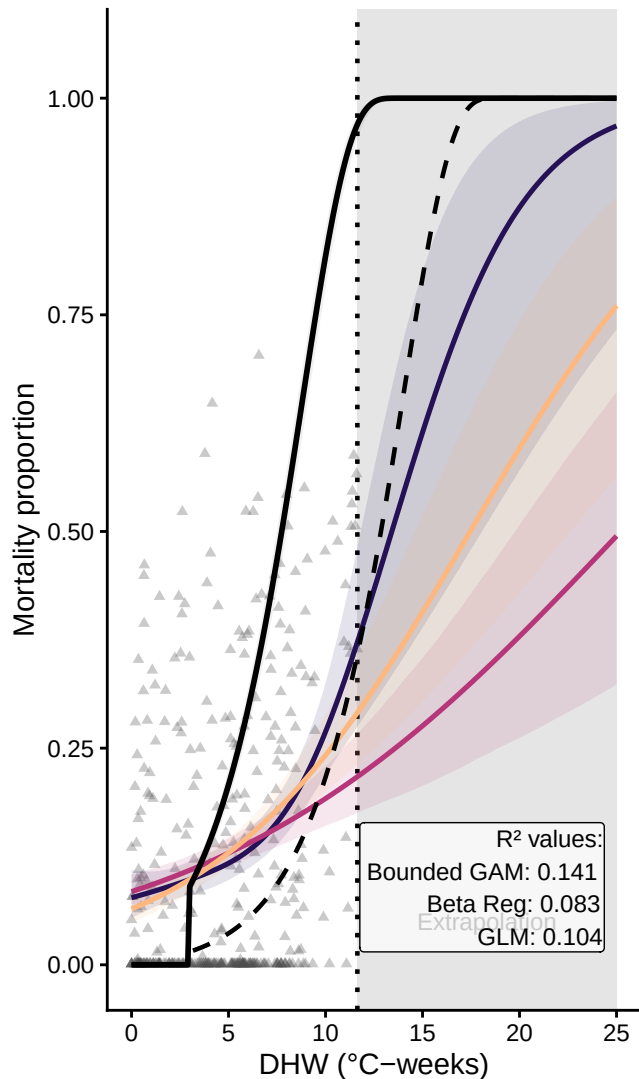




A) Hughes et al. (2018) — Single event (2016)



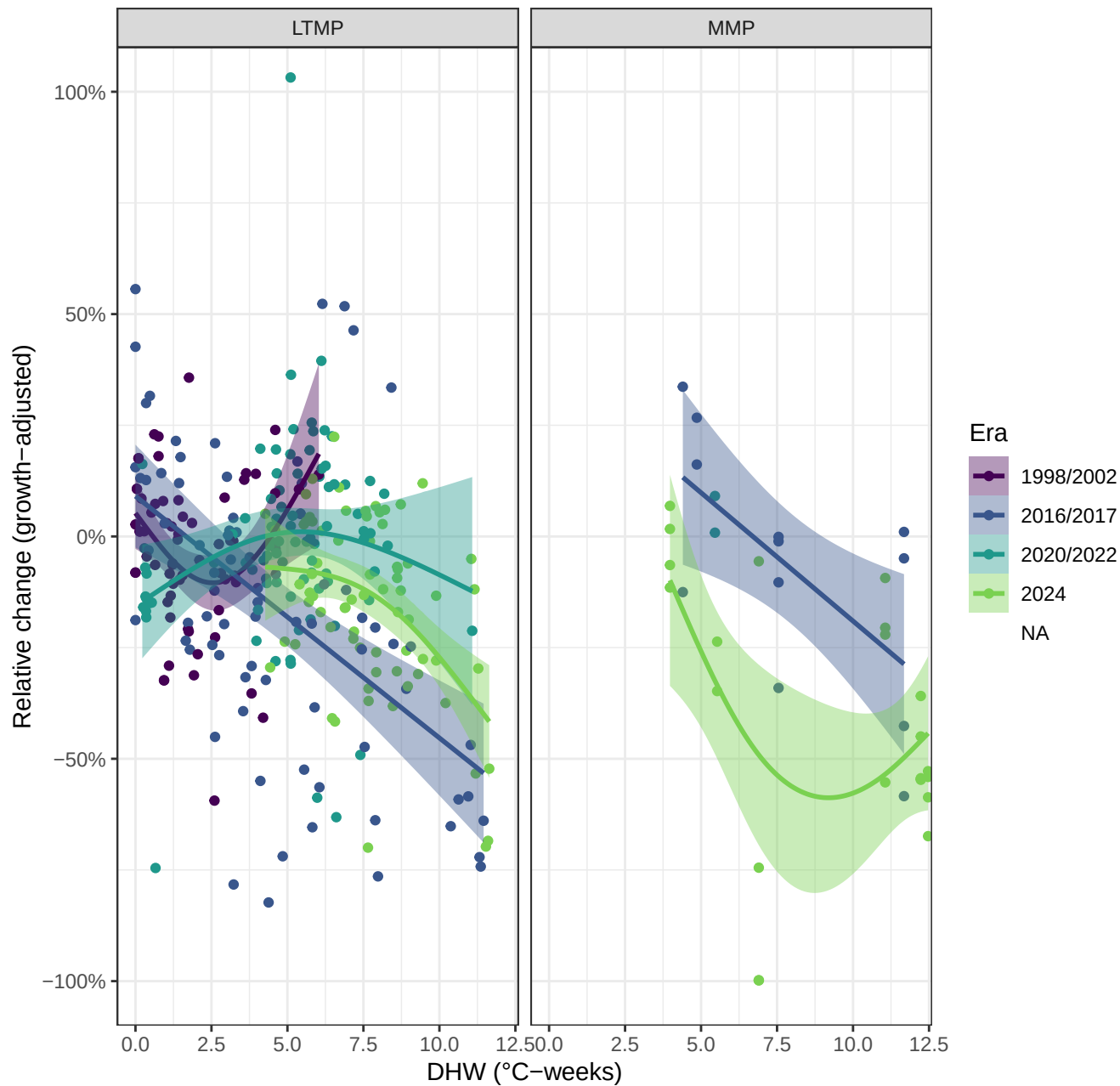
B) AIMS LTMP Manta Tow — Multi-event (2011-2015)

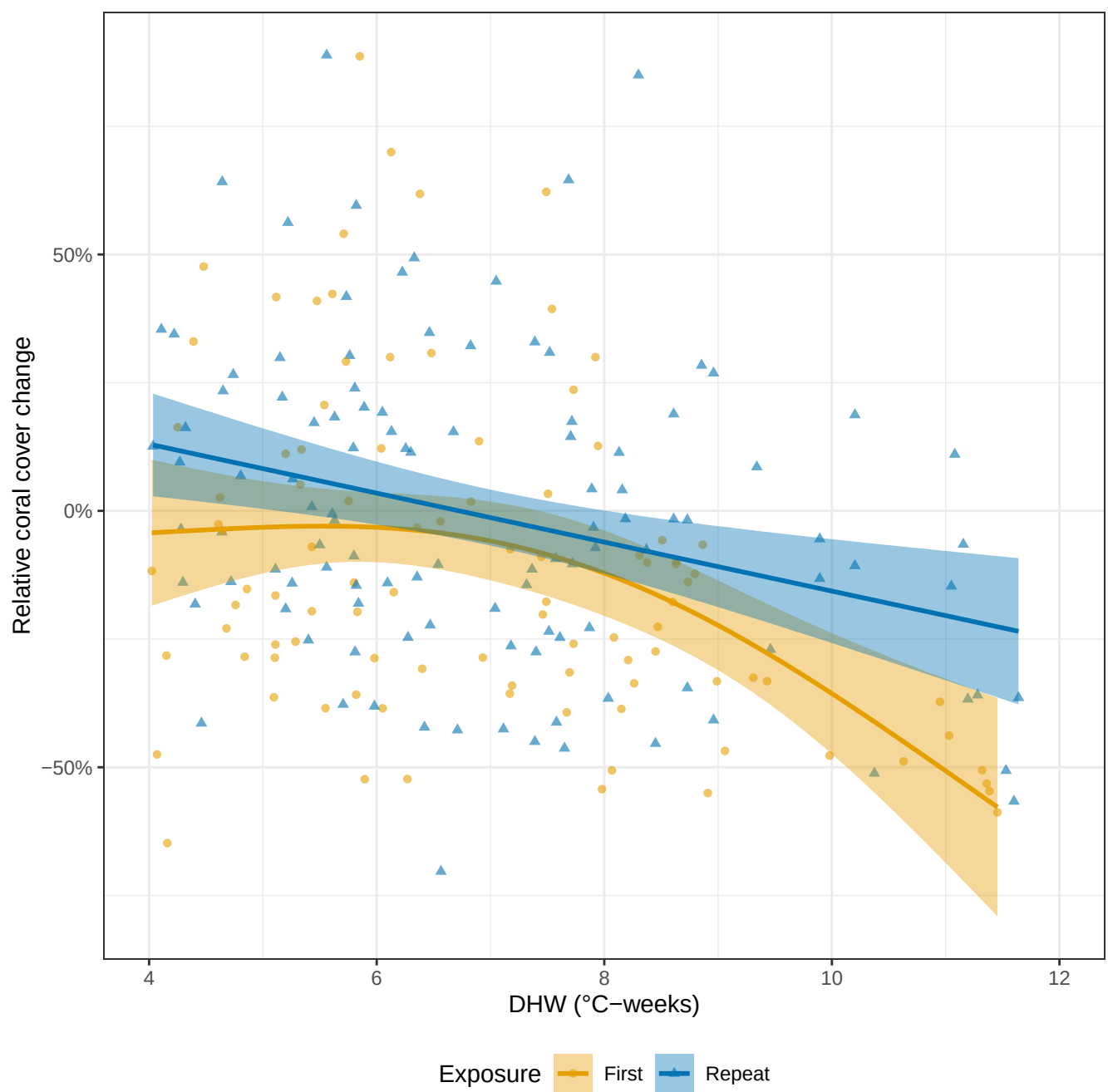


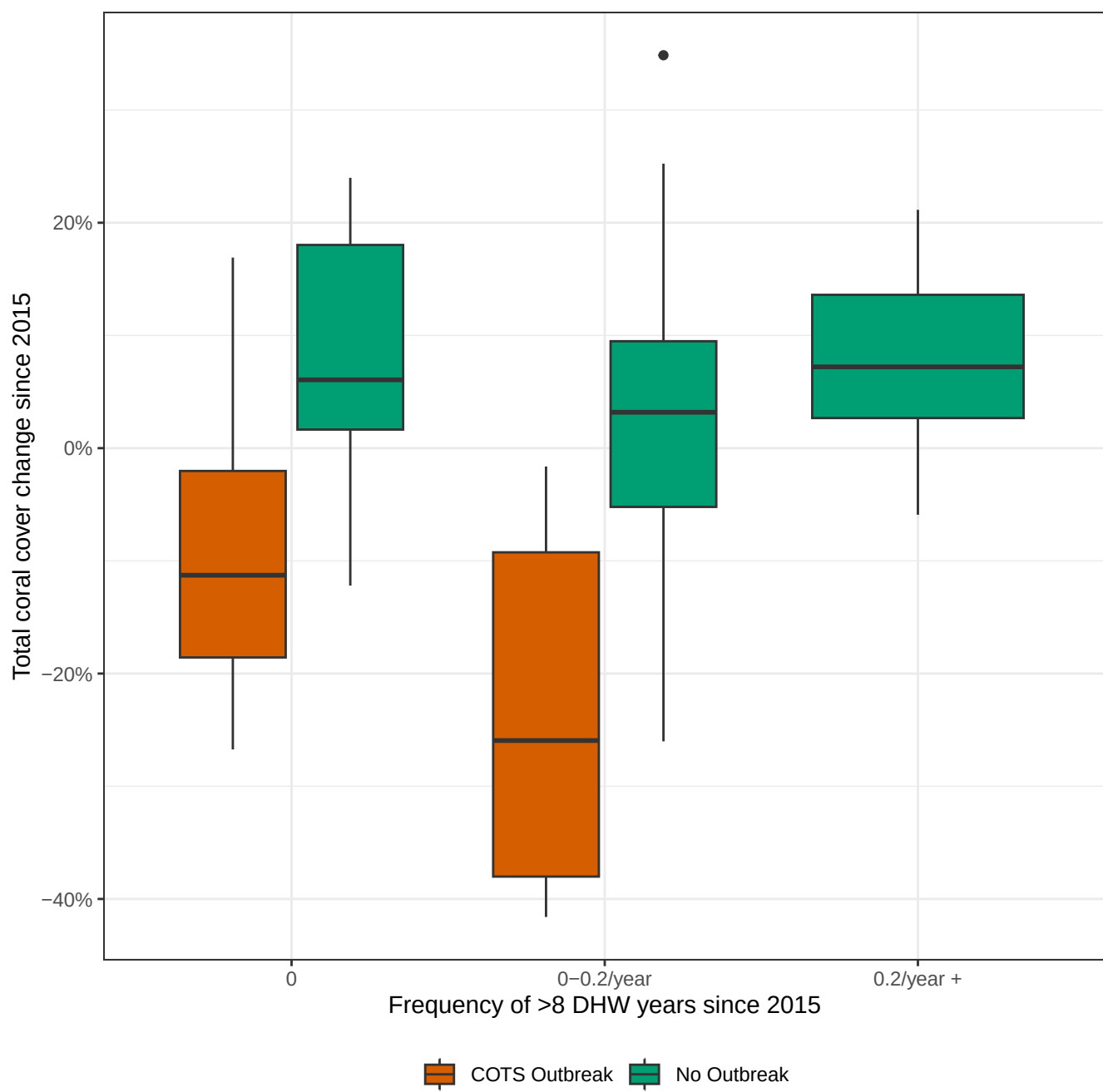
Bounded GAM (logit)
 Beta regression
 Quasi-binomial GLM

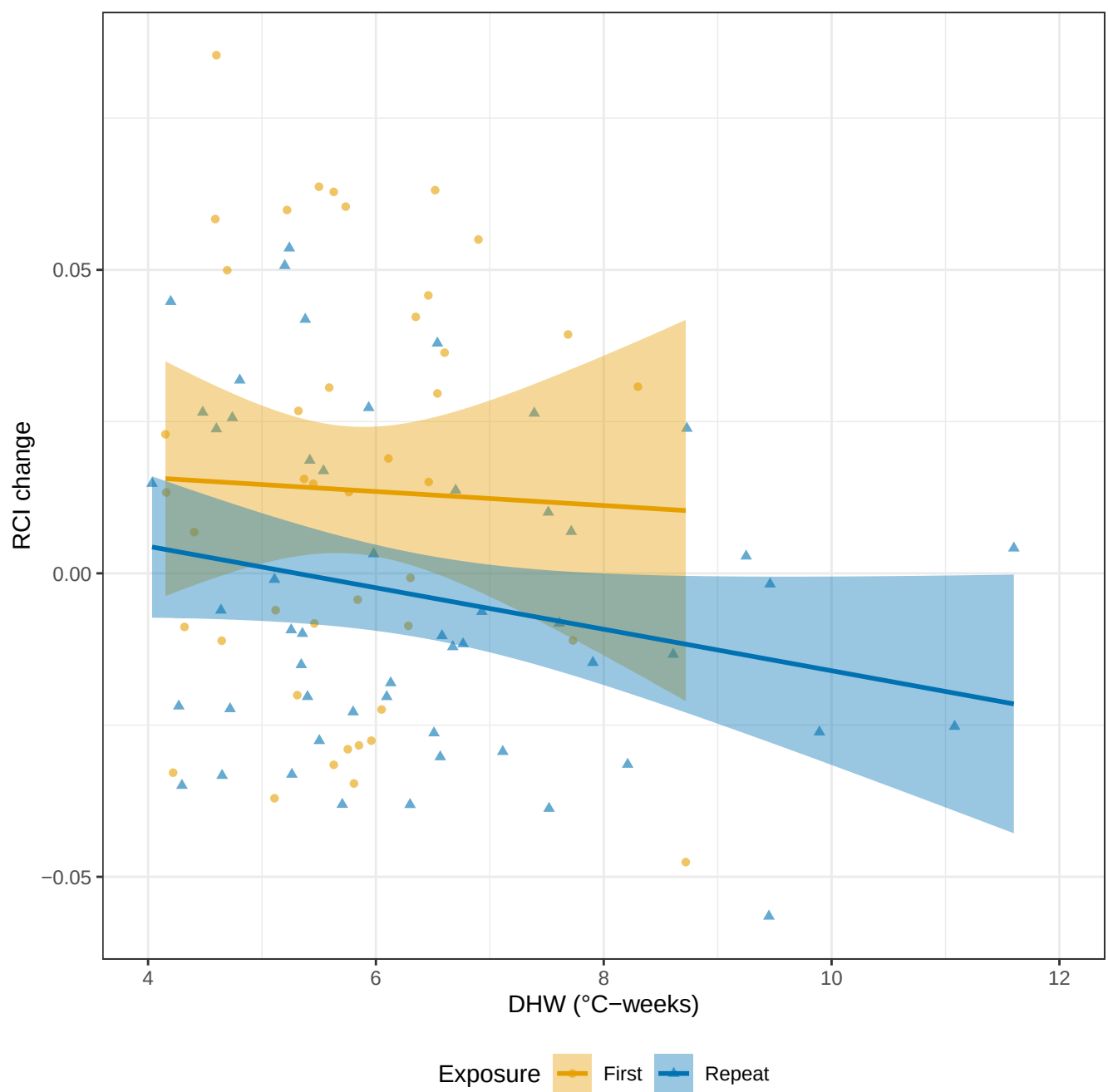
Note: AIMS mortality proportions are growth-corrected.
Reefs that gained cover are set to mortality ~ 0.

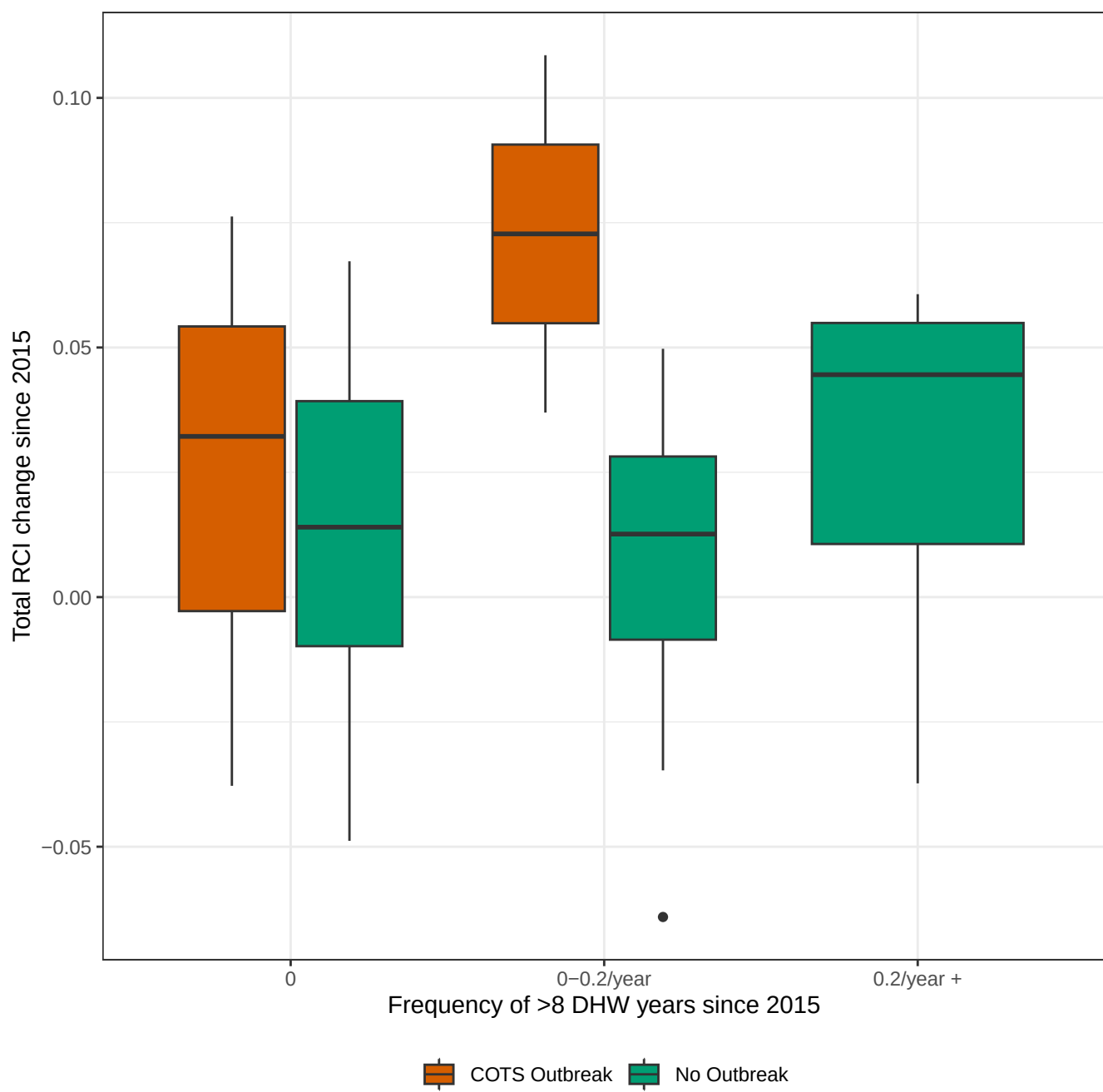
Benthic Transect: LTMP (Offshore) vs MMP (Inshore)

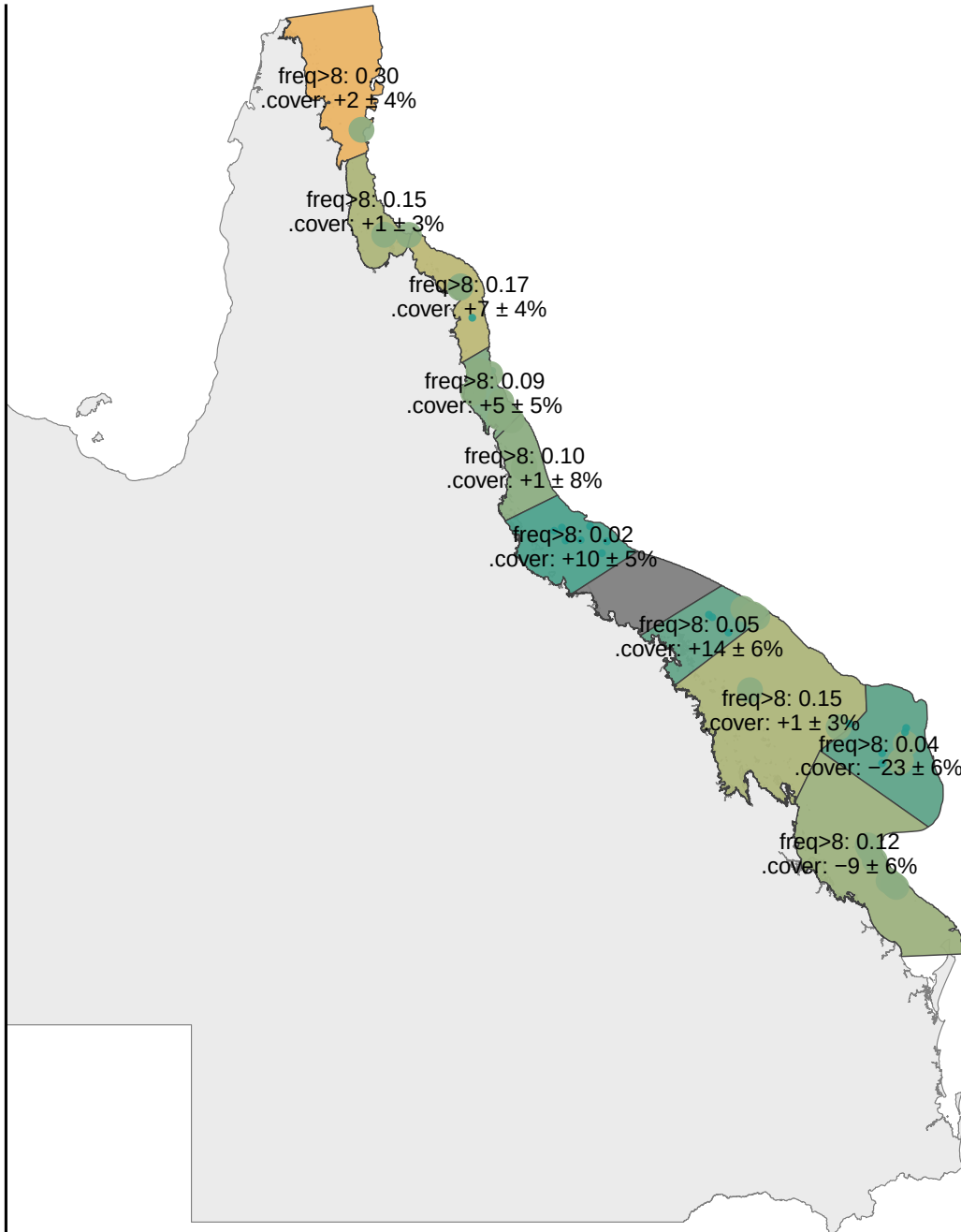




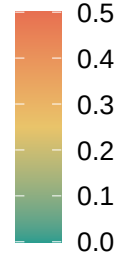








Freq >8 DHW



Spatial–Temporal IDW % Acropora Distribution (1985–2024)

